SOFTWARE REQUIREMENT SPECIFICATION [SRS]

E-PRASHIKSHAN

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Software Requirements Specification (SRS): E-Prashikshan 2.0 Learning Management System

1. Document Control & Cover Page

Project Name: e-Prashikshan 2.0 Document Title: Software Requirements Specification (SRS) Version: 1.0 Date: 15 May 2026

Document Control Block

Field	Value
Author	Technical Documentation Team
Status	Draft for Approval
Classification	Internal / Official
Prepared For	Administrative and Government Review

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2. Introduction

2.1 Purpose and Product Vision

The e-Prashikshan 2.0 platform is a scalable, role-driven digital learning ecosystem. It is architected to support the complete academic lifecycle—including course management, enrollment, assessment, and certification—within a secure, configurable framework. The primary purpose of this SRS is to provide a clear, auditable, and implementation-ready reference that supports high-quality governance and procurement transparency for all stakeholders.

2.2 Intended Audience

Government & Administrative Reviewers: For procurement and compliance oversight.

Product Owners: For strategic decision-making and institutional alignment.

System Architects & Developers: For technical implementation and code-level guidance.

Quality Assurance & UAT Teams: For validation, testing, and acceptance verification.

Security & Compliance Teams: For vulnerability assessment and data protection audits.

DevOps & Operations: For deployment, maintenance, and disaster recovery.

Training & Support: For user assistance and instructional documentation.

Analysis: Strategic Scope and Institutional Success

The strategic importance of this platform's vision lies in its ability to provide a standardized, transparent foundation for institutional learning. Defining the scope across 21 distinct modules is a critical architectural decision; it ensures that the platform is treated as an integrated enterprise resource rather than a collection of disparate tools. This prevents "scope creep" and ensures that vital operational components—such as grievance redressal and redistributable multi-instance capabilities—are woven into the core system fabric, rather than added as fragile third-party extensions.

2.3 Definitions and Acronyms

Term	Definition
LMS	Learning Management System
RBAC	Role-Based Access Control
API	Application Programming Interface
SSO	Single Sign-On
PII	Personally Identifiable Information
OTP	One-Time Password
SLA	Service Level Agreement
RTO	Recovery Time Objective (Max time to restore service)
RPO	Recovery Point Objective (Max allowable data loss age)
UI/UX	User Interface / User Experience
CRUD	Create, Read, Update, Delete operations
CSV	Comma-Separated Values (Data format for bulk imports)

Conclude by linking the high-level scope to the operational perspective of the system's role-based infrastructure.

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3. Overall Description

3.1 Product Perspective and Goals

e-Prashikshan 2.0 is a modular, service-oriented platform designed to handle complex instructional and administrative workflows. Its primary goals include:

Enabling end-to-end digital academic lifecycle management.

Ensuring secure, auditable governance through granular access controls.

Delivering measurable learner outcomes through institutional-grade analytics.

Providing an integration-ready architecture for future ecosystem expansion.

3.2 User Classes and Roles

User Class	Responsibility
Super Admin	Global system governance, configuration, and multi-tenant management.
Admin	Institution-level operations, user onboarding, and policy oversight.
Manager	Operational management aligned with specific institutional policies.
Course Creator	Instructional design, content authoring, and course structure mapping.
Teacher / Instructor	Content delivery, grading, attendance tracking, and student mentoring.
Non-editing Teacher	Instructional participation and grading without content modification rights.
Student / Learner	Participation in courses, assessment completion, and progress tracking.
Support / Reviewer	System audit, grievance management, and helpdesk operations.

Analysis: Decentralized Management and Permission Isolation

The role-based structure is the cornerstone of the platform's security and operational efficiency. By implementing a decentralized management model, the system allows individual institutions to manage their own users and courses without compromising the integrity of global system settings. This "So What?" is critical: it prevents a single point of failure where a local administrator could inadvertently affect global platform performance or access unauthorized multi-tenant data.

3.3 Assumptions and Dependencies

Infrastructure: High-availability (HA) cloud compute clusters with auto-scaling triggers based on CPU/Memory thresholds.

Services: Stable availability of SMTP for transactional emails and SMS provider APIs for 2FA/OTP.

Policy: Formalized inputs from institutions regarding grading rubrics, attendance thresholds, and refund policies.

Compliance: Continuous alignment of data protection patterns with evolving legal frameworks.

These user roles and goals are supported by the underlying system architecture, which provides the technical engine to enforce these boundaries.

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4. System Architecture

4.1 Logical and Deployment Architecture

The system utilizes a modular backend with tenant-aware data partitioning. This ensures that data from different institutions remain logically isolated within a shared infrastructure.

Logical Architecture: Three distinct interfaces:

Head: The administrative control plane for Super Admins.

Portal: The primary interface for instructors and learners.

Certification: A public-facing verification gateway.

Deployment Architecture: Cloud-hosted, stateless compute instances with automated process management (e.g., PM2 or Kubernetes) and controlled port exposure.

Analysis: API-First and Multi-Tenant Partitioning

The "API-first" design is a strategic mandate; it ensures future-proofing by treating the core logic as a set of secure services accessible to any frontend or third-party application. More importantly, tenant-aware data partitioning is the technical safeguard for multi-instance scalability. By isolating data at the persistence layer (using scoped identifiers or separate schemas), we ensure that an "Admin" from Institution A can never query records from Institution B, fulfilling a primary requirement for government-grade data sovereignty.

4.2 Technology Stack & Environment Requirements

Environments: Strictly isolated Development (Dev), Staging (UAT), and Production (Prod) tiers.

Stack Requirements: Modular backend services, secure RESTful APIs, and responsive frontend frameworks.

Configuration: 12-factor application methodology, managing configuration exclusively through environment variables.

This technical framework provides the engine for the functional modules described hereafter.

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5. Functional Requirements: Core Management Modules (1-4)

5.1 Module 1: User Management

Objective

Provide robust identity lifecycle management across all user categories.

Requirements

- Manual user account creation with role assignment for precise onboarding.
- Bulk user onboarding through CSV import with validation support for more than 1000 records per transaction.

- Secure self-registration functionality for eligible user categories.
- Secure login and logout mechanisms with session management.
- Role and permission management at both site and course levels for granular RBAC enforcement.
- User profile management with institutional metadata support.
- Secure password recovery and reset workflows to reduce support overhead.
- Login activity tracking including login time, device fingerprint, and session metadata for audit purposes.
- Account status management including Active, Suspended, and Locked states for immediate access control.
- Complete CRUD lifecycle management for all user identities to ensure full identity governance.

5.2 Module 2: Course Management

Objective

Enable complete academic course lifecycle management from creation to archival.

Requirements

- Manual and bulk course creation support for rapid catalog expansion.
- Category and sub-category hierarchy management for logical content organization.
- Course scheduling with configurable start and end windows for temporal enrollment control.
- Searchable course catalog for efficient course discovery.
- Structured content hierarchy management including sections, topics, and items for scalable content delivery.
- Reusable course templates to accelerate course authoring and deployment.
- Multimedia upload and external content linking support for rich instructional media.
- Drag-and-drop content block ordering for intuitive course design.
- Progressive content unlock and conditional visibility management for controlled learning paths.
- Course completion criteria configuration with integrated progress tracking.
- Course-level role assignment and permission management for localized governance.
- Batch creation and allocation support for efficient academic resource planning.
- Course activity logging and participation reporting for detailed audit tracking.
- Prerequisite mapping between courses to maintain validated learning sequences.

5.3 Module 3: Certificate Generation

Objective

Deliver trusted, automated, and verifiable digital certification workflows.

Requirements

- Certificate template design and management for branded digital credentialing.
- Criteria-based certificate generation using completion status, grades, and attendance records.
- Automatic certificate generation for eligible learners to reduce administrative effort.
- Certificate PDF rendering and download functionality for persistent digital records.
- Secure unique code and QR code embedding for tamper-proof credential verification.
- Public certificate verification using unique code or QR scan for external trust validation.
- Administrative certificate issuance tracking for auditable certification records.
- Certificate resend and reissue workflows to ensure service reliability and accessibility.

5.4 Module 4: Coupons and Enrollment Management

Objective

Enable flexible enrollment pathways with controlled access and promotional logic.

Requirements

- Manual enrollment support for authorized staff to ensure controlled and high-touch user placement.
- Self-enrollment functionality through direct links and coupon-based access mechanisms.
- Coupon generation and discount control management to maintain fiscal and promotional policy compliance.
- Bulk enrollment import through CSV files for efficient large-scale onboarding operations.
- Enrollment approval workflows for controlled and policy-driven access governance.
- Group-wise and batch-wise enrollment controls for organized cohort management.
- Enrollment key expiry settings and usage limit restrictions for secure and limited-access distribution.
- Automatic role assignment during enrollment for immediate privilege allocation and access provisioning.
- Enrollment analytics and audit trail management for complete enrollment lifecycle transparency.

Analysis: Credibility and Fraud Prevention

Automated certificate generation combined with QR-based public verification acts as a critical fraud prevention mechanism within institutional and government environments. By enabling secure, publicly verifiable credentials through cryptographically linked QR codes and unique verification identifiers, the platform significantly reduces the possibility of forged or manipulated certificates. This strengthens institutional trust, enhances authenticity validation, and improves the credibility of the issuing authority among external stakeholders, employers, and regulatory bodies.

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6. Functional Requirements: Academic Integrity & Feedback (Modules 5-7)

6.1 Module 5: Assessment

Objective

Deliver comprehensive evaluation with robust grading and academic integrity.

Requirements

- Support for multiple assessment types including MCQ, True/False, Short Answer, Essay, and Matching questions to enable diverse evaluation methods.
- Question bank creation, categorization, tagging, and reusable assessment content management for long-term content sustainability.
- Configurable assessment settings including attempt limits, timers, passing scores, and negative marking policies for rigorous examination environments.
- Randomization of question sequences and answer option ordering to reduce collusion and maintain academic integrity.
- Automatic grading support for objective question types such as MCQs and True/False assessments to provide instant learner feedback.
- Manual grading workflows for subjective and descriptive responses to enable qualitative evaluation.
- Assignment submission and grading workflows for practical and document-based skill validation.
- Rubric-based evaluation mechanisms to ensure consistent and standardized grading quality.
- Integration of assessment outcomes with the centralized gradebook for unified academic record management.
- Learner performance analytics and reporting tools to identify learning gaps and support targeted remediation strategies.

6.2 Module 6: Feedback Management

Objective

Collect structured and actionable feedback for continuous improvement.

Requirements

- Customizable feedback template creation and management for standardized data collection processes.
- Support for multiple feedback question formats including rating scales, descriptive responses, MCQs, and surveys for granular sentiment analysis.

- Conditional feedback logic to present relevant questions based on user responses and contextual workflows.
- Schedule-based feedback windows to enable time-bound and context-aware feedback collection.
- Anonymous and identified feedback submission modes to encourage honest and transparent responses.
- Role-based access control for feedback forms to ensure scoped and targeted response collection.
- Feedback analytics, reporting, and export functionality for actionable quality assessment and decision-making.
- Feedback-linked completion policies to enforce higher response participation and compliance rates.

6.3 Module 7: Attendance Management

Objective

Provide reliable and auditable attendance operations across modalities.

Requirements

- Manual and bulk attendance marking support for flexible and scalable attendance management.
- Attendance status management including Present, Absent, Late, and Leave for accurate participation tracking.
- Session-date and calendar-based attendance recording for structured temporal monitoring.
- QR-based attendance capture functionality for secure and contactless authentication workflows.
- Attendance analytics and trend reporting tools to identify absenteeism patterns and learner disengagement.
- Attendance export support in common formats for external audits and institutional reporting Requirements.
- Attendance contribution integration with grading systems for compliance-driven academic evaluation.
- Student self-service attendance view to promote transparency and self-accountability among learners.

Analysis: Integrity and Administrative Efficiency

The integration of assessment randomization and automated grading mechanisms addresses both academic integrity concerns and operational efficiency challenges. Randomized question sequencing and answer option shuffling create individualized assessment environments, significantly reducing the effectiveness of collusion and shared answer patterns. Simultaneously, automated grading workflows provide immediate learner feedback, improve assessment turnaround time, and substantially reduce manual evaluation overhead for instructors and administrative staff.

Analysis: Integrity and Administrative Efficiency

The implementation of randomization (FR-ASM-04) and automated grading (FR-ASM-05) addresses the dual challenge of academic integrity and administrative burden. By programmatically shuffling test vectors, the system creates a unique environment for every student, drastically reducing the efficacy of shared answer keys. Simultaneously, auto-grading provides instantaneous feedback, which is statistically proven to improve learning outcomes, while freeing instructors from hundreds of hours of manual marking.

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7. Functional Requirements: Operational Intelligence (Modules 8-11)

7.1 Module 8: Dashboard

Objective

Provide role-specific operational intelligence for rapid decision-making.

Requirements

- Administrative dashboard with system-wide KPIs and operational metrics for centralized platform monitoring and governance.
- Teacher dashboard providing class performance insights, learner progress tracking, and instructional analytics for improved teaching effectiveness.
- Learner dashboard displaying course progress, upcoming deadlines, achievements, and personalized learning milestones.
- Badge and achievement indicator system to encourage gamified engagement and learner motivation.
- Configurable leaderboard functionality to promote peer-driven competition and participation.
- Trend visualization widgets for enrollment, completion rates, and performance analytics to support predictive operational decision-making.

7.2 Module 9: Communication

Objective

Enable seamless institutional communication and engagement.

Requirements

- Private messaging functionality among authorized users to support peer, mentor, and institutional collaboration.
- Global, course-level, and batch-specific announcement management for targeted information dissemination.
- Calendar event scheduling and reminder notifications to improve temporal awareness and activity coordination.
- Email notification support for critical events, updates, and workflow triggers to ensure off-platform engagement.
- Configurable communication preferences allowing users to personalize notification and alert settings.
- Chat systems and asynchronous discussion pathways for continuous collaboration and persistent knowledge exchange.

- Communication audit logs to maintain traceable and defensible records of institutional communication activities.

7.3 Module 10: Payment and Billing

Objective

Support sustainable operations through transparent financial workflows.

Requirements

- Plan and subscription management capabilities to support flexible commercial and institutional pricing models.
- Payment gateway integration for secure real-time transaction processing and seamless revenue collection.
- Multi-mode payment support including various digital and banking channels for broader financial accessibility.
- Coupon and discount policy management for dynamic pricing governance and promotional workflows.
- Automatic digital receipt generation for successful transactions to maintain transparent fiscal records.
- Tax calculation and accounting metadata support for regulatory and financial compliance Requirements.
- Refund management workflows with policy-driven controls for secure financial reversal handling.
- Billing and payment analytics dashboards for revenue monitoring and institutional ROI visibility.

7.4 Module 11: Content Management

Objective

Provide centralized, secure, and reusable academic content management.

Requirements

- Centralized content repository for secure storage and unified management of institutional learning assets.
- Multimedia and rich-content support including videos, documents, presentations, and interactive learning materials.
- Content search, filtering, and tagging functionality for rapid and efficient academic asset discovery.
- Role-based content sharing and access control mechanisms to protect intellectual property and manage permissions.
- Version awareness and update tracking to maintain content lifecycle integrity and revision transparency.
- File type validation, upload size restrictions, and security policy enforcement for repository governance and hygiene.

Analysis: Operational Agility via KPIs

The integration of trend visualization widgets and achievement indicators transforms the platform from a passive information repository into an active operational intelligence system. Institutional KPIs such as enrollment trends, course completion rates, learner participation, and engagement metrics act as early-warning indicators for administrators and decision-makers. Sudden declines in activity or performance can trigger proactive investigation and intervention, enabling institutions to shift from reactive management approaches to predictive and data-driven operational governance.

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8. Functional Requirements: Security, Access, & Inclusivity (Modules 12-14)

8.1 Module 12: Security and Compliance

Objective

Ensure secure, compliant, and auditable platform operations.

Requirements

- Secure password hashing and password policy enforcement to protect user identities and credentials.
- Account lockout mechanisms after repeated failed login attempts to mitigate brute-force attacks.
- Role-Based Access Control (RBAC) and permission isolation for zero-trust access governance.
- Two-Factor Authentication (2FA) and OTP-based authentication workflows for enhanced account security.
- Personally Identifiable Information (PII) masking and controlled field visibility to support privacy-by-design principles.
- Comprehensive audit logging for all critical user and administrative actions to ensure forensic readiness.
- Request throttling, rate limiting, and intrusion prevention safeguards to maintain service availability and protect against denial-of-service attacks.
- Secure session and token management mechanisms to prevent session hijacking and unauthorized access.
- Periodic security hardening and compliance validation processes for continuous risk management and governance assurance.

8.2 Module 13: Mobile Learning

Objective

Enable effective learning access on mobile devices.

Requirements

- Responsive user interfaces optimized for mobile browsers to support device-agnostic learning experiences.
- Mobile-first interaction patterns with touch-friendly navigation and adaptive layouts for intuitive usability.
- Push notification support for academic events, reminders, and real-time learner communication.
- Selective offline-ready content access strategies to support learning continuity in low-connectivity environments.
- Native application channel readiness to support seamless mobile ecosystem expansion and future scalability.

8.3 Module 14: Multilingual Support

Objective

Provide inclusive language accessibility for diverse users.

Requirements

- User-selectable interface language settings to improve demographic inclusivity and accessibility.
- Localization support for core labels, workflows, and system interactions to provide native-language user experiences.
- Multilingual content management capabilities for equal educational access across language groups.
- Persistent storage of user language preferences to ensure consistent multilingual session experiences.
- Extensible translation workflows to support future language additions and global scalability.

Analysis: Inclusivity and Adoption

The combination of multilingual accessibility and mobile-first interaction design is a critical factor in ensuring widespread platform adoption across geographically and linguistically diverse user groups. In environments with varying levels of digital literacy and infrastructure maturity, providing localized mobile experiences significantly lowers adoption barriers. Together, these capabilities ensure that technology functions as an inclusive educational enabler rather than a limiting constraint.

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9. Functional Requirements: Data Stewardship & Interoperability (Modules 15–18)

9.1 Module 15: Tracking and Reporting

Objective

Provide measurable evidence of progress and outcomes.

Requirements

- Tracking of key user activity events to maintain granular operational and academic audit trails.
- Granular course progress tracking for real-time learner completion and engagement visibility.
- Grade and performance reporting mechanisms to measure academic outcomes and learner achievement.
- Attendance and engagement analytics for identifying participation trends and behavioral insights.
- Export support for reports in formats such as PDF and CSV to ensure portable and shareable evidence records.
- Configurable alert systems for predefined academic and operational risk thresholds to enable early intervention strategies.

9.2 Module 16: Backup, Restore, and Archiving

Objective

Ensure business continuity and long-term data stewardship.

Requirements

- Automated backup scheduling to minimize the risk of institutional data loss.
- Manual on-demand backup creation for point-in-time recovery assurance.
- Selective restore functionality by domain or dataset for targeted recovery operations.
- Archive policy management for inactive or historical data to optimize storage utilization and retention governance.
- Backup integrity validation processes to ensure reliable recovery and restoration readiness.
- Disaster Recovery (DR) procedures with defined Recovery Time Objective (RTO) and Recovery Point Objective (RPO) targets for institutional resilience.

9.3 Module 17: API and Integration

Objective

Enable standardized interoperability with external systems.

Requirements

- Secure REST API support for core system domains to enable ecosystem connectivity and interoperability.
- API documentation and schema reference management to improve developer efficiency and integration accuracy.
- Token-based API authentication workflows for secure zero-trust integrations.
- Third-party institutional integration support for external systems, services, and operational workflows.
- API versioning and deprecation management to ensure long-term integration stability and backward compatibility.
- Integration monitoring and error-handling mechanisms to maintain reliable and traceable data exchanges.

9.4 Module 18: Online Meeting

Objective

Support live synchronous learning and collaboration.

Requirements

- Secure video conferencing support for virtual classroom delivery and live instructional sessions.
- Audio conferencing functionality optimized for low-bandwidth environments to ensure accessibility.
- Collaborative whiteboard tools for synchronous brainstorming and interactive teaching sessions.
- Screen-sharing capabilities for demonstrations, presentations, and instructional clarity.
- Breakout room management for peer collaboration and group-based learning activities.
- In-session chat and polling tools to support real-time learner interaction and dynamic feedback collection.

Analysis: Disaster Recovery and Business Continuity

The implementation of clearly defined Recovery Time Objectives (RTO) and Recovery Point Objectives (RPO) is essential for ensuring institutional business continuity during catastrophic infrastructure or service failures. Automated and integrity-validated backup mechanisms guarantee the long-term protection of academic records, certifications, and operational data. By prioritizing rapid recovery and resilient restoration strategies, the platform safeguards legally significant educational records and minimizes operational disruption during crisis scenarios.

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10. Functional Requirements: Scaling & Redressal (Modules 19-21)

10.1 Module 19: SEO

Objective

Improve discoverability and public content indexing.

Requirements

- Metadata management support for public-facing pages to improve search engine visibility and content discoverability.
- Structured data markup implementation to enable rich snippets and enhanced search result presentation.
- Integration with analytics platforms and webmaster tools for traffic monitoring and search performance insights.
- Sitemap generation and indexing control mechanisms to ensure efficient search engine crawling and content indexing.
- Multilingual SEO metadata support to enhance global organic reach across diverse language audiences.

10.2 Module 20: Redistributable Multi-Instance Module

Objective

Enable scalable, policy-controlled deployment of multiple institutional instances.

Requirements

- Multi-tenant isolation architecture to ensure complete separation of institutional data and prevent cross-instance data leakage.
- Instance-level governance controls to provide operational autonomy for individual institutions or sub-entities.
- Module enablement and configuration management at the instance level for customizable feature availability.
- Instance provisioning templates for standardized, rapid, and repeatable node deployment processes.
- Branded and domain-specific customization support to maintain institutional identity and localized user experience.
- Per-instance reporting and analytics capabilities for localized operational intelligence and decision-making.

- Scalable deployment architecture designed to support horizontal platform growth and large-scale institutional expansion.

10.3 Module 21: Grievance Redressal

Objective

Provide transparent and accountable complaint management.

Requirements

- Complaint submission functionality with category selection for structured issue registration and tracking.
- Automatic generation of unique ticket numbers to ensure end-to-end grievance traceability.
- Complaint lifecycle status tracking including submitted, under review, resolved, and closed states for complete user transparency.
- Assignment of grievances to relevant departments or authorities for accountable issue resolution.
- Threaded communication and response management to maintain documented interaction history between users and authorities.
- Resolution feedback and rating mechanisms for validating service quality and user satisfaction.
- Grievance analytics and trend reporting tools to identify recurring operational or systemic issues.
- SLA-based escalation workflows and automated escalation rules to enforce response timelines and prevent unresolved ticket stagnation.

Analysis: Multi-Instance Scaling and Redressal

The multi-instance architecture serves as a foundational mechanism for scalable institutional deployment, enabling rapid onboarding of new organizations while maintaining centralized governance and policy enforcement. Complementing this architecture, the grievance management module ensures that accountability mechanisms scale proportionally with platform expansion. Automated SLA enforcement and escalation workflows prevent complaint stagnation, support timely issue resolution, and strengthen institutional trust through transparent service management practices.

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11. External Interface Requirements

11.1 Interface Definitions

User Interface: A unified, responsive design language across Head, Portal, and Certification interfaces. Consistent interaction patterns are mandated to reduce cognitive load and minimize institutional training costs.

API Interface: Standardized RESTful patterns using JSON for data exchange. Every endpoint requires token-based (JWT) authentication and detailed version-friendly contracts.

Communication Interface: Integration with SMTP for system-triggered emails and SMS Gateway for transactional OTP alerts.

Data Interface: Secure database interactions managed via an Object-Relational Mapping (ORM) layer with strict tenancy-aware filters to prevent unauthorized cross-tenant data access.

The effectiveness of these interfaces is measured by their ability to provide consistent UX/UI behavior, ensuring that a user transitioning from mobile to desktop experiences zero friction in workflow completion.

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12. Non-Functional Requirements (NFRs)

ID	Category	Requirement Description
NFR-PERF-01	Performance	Deliver response times < 2s for core CRUD operations under peak load.
NFR-AVL-01	Availability	Support High Availability (HA) with an uptime target of 99.9%.
NFR-SEC-01	Security	Enforce AES-256 encryption for data at rest and TLS 1.3 for data in transit.
NFR-SCL-01	Scalability	Horizontal scaling of compute nodes to support >10k concurrent sessions.
NFR-USE-01	Usability	Core workflows must be navigable with <3 clicks for non-technical users.
NFR-MNT-01	Maintainability	Modular code structure with standardized logging for rapid triage.
NFR-CMP-02	Compliance	Maintain security and audit controls for government review readiness.

Analysis: Performance and Institutional Trust

Enterprise-grade institutional trust is built on reliability. If the system fails to maintain "Acceptable Response Times" during high-stakes assessment windows, the resulting loss of user confidence can jeopardize the entire project. High Availability is not an option; it is the fundamental requirement for a platform that serves as the official academic record.

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13. Data & Security Requirements

13.1 Core Data Domains

The system manages: User Identity, Course Hierarchy, Enrollment/Coupons, Assessments/Grades, Attendance/Feedback, Certificates, Billing Metadata, and Grievance records.

13.2 Security Governance

Security is enforced through RBAC and OTP-hardened authentication. PII handling is strictly governed by data masking and field-level visibility controls.

13.3 Audit and Logging Requirements

AR-01: System shall maintain immutable action logs for all administrative workflows.

AR-02: Searchable operational logs must be accessible for technical troubleshooting.

AR-03: All authentication and authorization events (success/failure) must be captured.

AR-04: Provide real-time monitoring hooks for uptime and security incidents.

AR-05: Support traceable incident response workflows via logged metadata.

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14. Workflow Requirements

14.1 Enrollment Workflow

Selection: User selects a course or initiates a coupon-based entry .

Validation: System validates eligibility, discount validity, or payment status .

Approval: System or Admin approves the enrollment based on institution-level policy .

Activation: The system assigns the appropriate course-level role and adds the user to the attendance roster .

14.2 Certification Workflow

Evaluation: Learner completes course Requirements and meets defined grade/attendance thresholds .

Trigger: System initiates the auto-generation service .

Generation: PDF is rendered with a unique QR code and cryptographic signature .

Issuance: The certificate is pushed to the learner dashboard and logged in the admin issuance report .

14.3 Grievance Escalation

Submission: User submits a ticket with a defined category .

Tracking: System generates a unique ticket ID for traceability .

Response: The assigned authority provides a threaded update within the SLA window .

Escalation: If the ticket remains unresolved past the threshold, the system auto-escalates to the next authority tier.

Analysis: Workflow Accountability

By linking automated escalation rules (FR-GRV-08) directly to the ticket lifecycle, the platform ensures that no grievance remains unaddressed. This enforces institutional accountability and ensures that service-level agreements are not just targets, but operational realities.

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15. Integration & Compliance Requirements

Gateways: Secure integrations for Payment (PCI-DSS compliant), SMS (Transactional), and Video Conferencing (WebRTC/Zoom/Teams).

Audit Readiness: (NFR-CMP-02) ensures that the platform is perpetually ready for external security audits, maintaining long-term legal standing for institutional procurement.

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16. Quality Assurance & Acceptance Criteria

16.1 Verification Framework

The QA process will involve: Unit testing for service logic, Integration testing for API workflows, User Acceptance Testing (UAT), Performance load testing, and Security vulnerability scanning.

16.2 Acceptance Criteria Framework

Functionality: All 21 modules pass functional verification against FR-IDs.

Governance: RBAC policies are strictly enforced without privilege escalation.

Resilience: Negative and edge-case testing (e.g., partial payments, concurrent session limits) is documented and passed.

Traceability: Every critical action generates a corresponding, unalterable audit trace.

User Success: Documented evidence that UAT scenarios were completed successfully by end-user proxies.

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17. Deployment, DevOps, & Risks

17.1 Implementation Priority Roadmap

Phase 1: Security hardening, identity management, and core course stability.

Phase 2: Assessment engine depth, certificate reliability, and dashboard intelligence.

Phase 3: Multi-channel communications, payment expansion, and reporting enhancements.

Phase 4: Backup/DR validation, integration maturity, and multi-instance scaling.

Phase 5: Advanced mobile, SEO, live meeting ecosystem, and performance optimization.

Analysis: Phase 1 Security Priority

Security hardening is mandated in Phase 1 to prevent the propagation of insecure session handling or data access patterns into the complex multi-tenant logic of Phase 4. By establishing a "hardened core" first, the platform ensures that subsequent modules inherit a robust security posture, reducing the technical debt and risk profiles for the entire platform.

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18. Appendices and Sign-Off

18.1 Glossary

PII: Personally Identifiable Information.

RTO/RPO: Recovery Time Objective / Recovery Point Objective.

RBAC: Role-Based Access Control.

Head: The Administrative backend control interface.

18.2 Sign-Off Section

e-Prashikshan 2.0 is an enterprise-ready framework designed for measurable academic delivery and secure institutional governance. By implementing the 21 modules detailed in this SRS, the platform ensures sustainable growth, administrative accountability, and a high-trust digital learning environment.